

21EIE552T – CYBERSECURITY IN INDUSTRIAL AUTOMATION

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Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1 :	<i>Introduce the basic knowledge in industrial automation and control systems</i>
CLR-2 :	<i>Know the basic concepts of information system security for industrial control systems</i>
CLR-3 :	<i>Understand the difference between IACS and IT paradigms</i>
CLR-4 :	<i>Impart the adequate information about risk management for IACS</i>
CLR-5 :	<i>Introduce the security methodologies and approaches for IACS</i>
CLR-6 :	<i>Understand the need of cybersecurity for different industrial applications</i>

Course Outcomes (CO):	At the end of this course, learners will be able to:
CO-1 :	Describe the basic concepts of industrial automation and control systems
CO-2 :	Illustrate any application with add-on security features in industrial control systems
CO-3 :	Select the software with IT paradigms for IACS
CO-4 :	Analyze the risk management approach for IACS
CO-5 :	Examine the security methodologies for industrial systems
CO-6 :	Design cybersecurity resilient algorithm for selective applications

Duration (hour)		Industrial Automation and Control System(IACS) Fundamental Concepts
		12
S-1	SLO-1	IACS - Introduction
	SLO-2	SCADA Systems
S-2	SLO-1	Distributed Control Systems
	SLO-2	Safety Instrumented Systems
S-3	SLO-1	IACS - Protocol
	SLO-2	OSI Model
S-4	SLO-1	Tutorial: 1 – Distributed Control
	SLO-2	
S-5	SLO-1	TCP/IP Model
	SLO-2	OPC for Process Control
S-6	SLO-1	TCP, DNP3 Protocol

	SLO-2	Utility Communication Architecture
S-7	SLO-1	Profibus
	SLO-2	Controller Area Network
S-8	SLO-1	Tutorial: 2 – Process Control
	SLO-2	
S-9	SLO-1	Ethernet/IP
	SLO-2	Open safety Protocol
S-10	SLO-1	Issues in IACS security
	SLO-2	Information Security
S-11	SLO-1	<i>Approaches of Information Security</i>
	SLO-2	Applications
S-12	SLO-1	Tutorial: 3 – ICT security
	SLO-2	

- What comes to your mind when you hear the words

Industrial

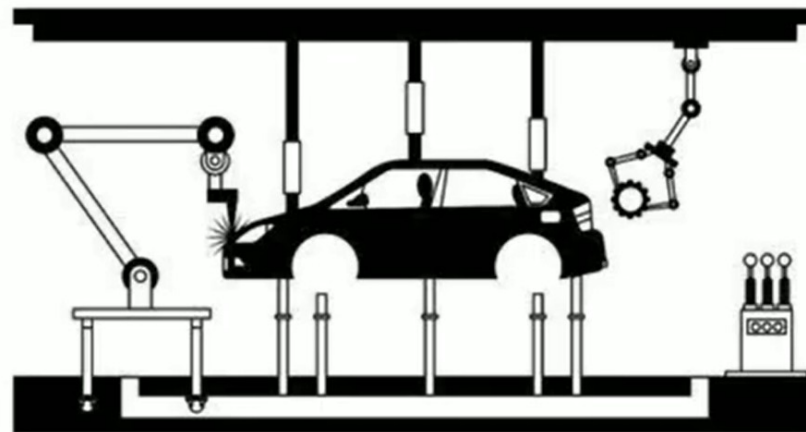
&

Automation



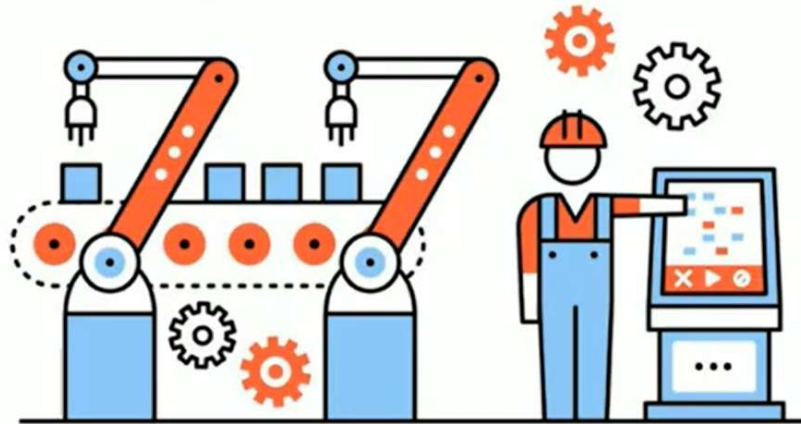
Introduction to Automation

Automation is the technology by which a process is performed with minimum human efforts



What is industrial automation..?

Industrial automation is the use of control system, such as computer or robots for handling different processes & machineries in an industry to replace a human being.



Use of Automation:

1. Automotive manufacturing Industries 

2. Food industries 

3. Agriculture Industries 

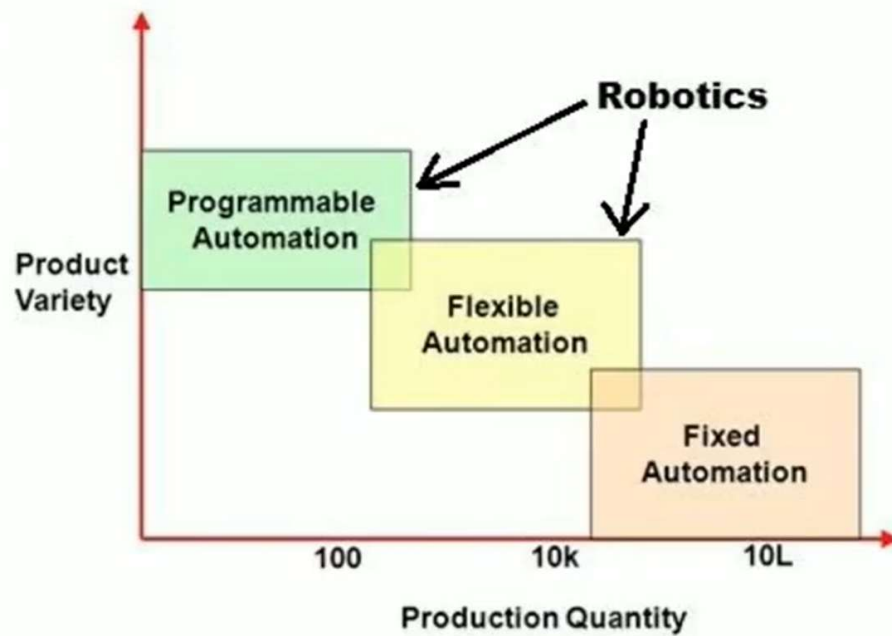
4. Packaging Industries 

5. Medical Industries 

Types of Automation

1. Fixed Automation (Hard Automation)
2. Programmable Automation (Soft Automation)
3. Flexible Automation

Types of Automation



Advantages:

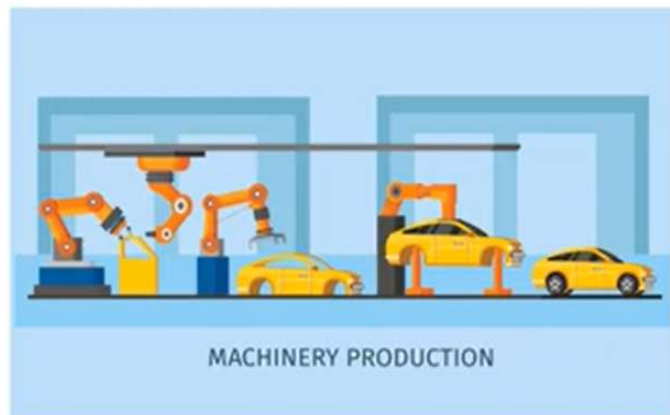
1. To increase the productivity.
2. To Improve the quality of products.
3. To increase the safe work environment.
4. To make better manufacturing.

Disadvantages:

1. Initial investment cost is very high.
2. Increase in Unemployment.

Industrial Automation

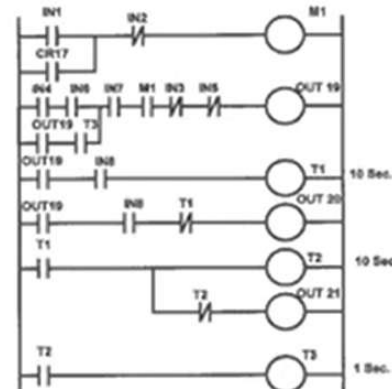
- The course contents are distinctively subdivided into:
- **Concepts related to Industrial Automation and Manufacturing Operations**
 - Need for Industrial Automation
 - Production Systems
 - Automation Strategies
 - Various Kinds of Manufacturing Operations
 - Manufacturing Economics





Industrial Automation

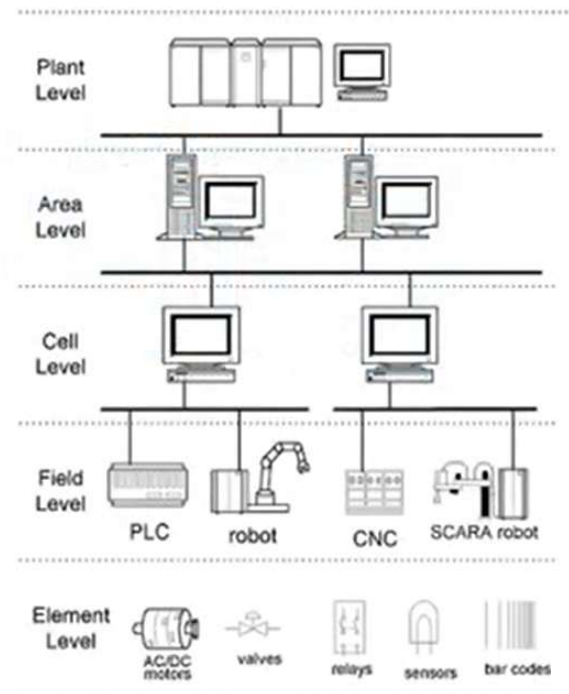
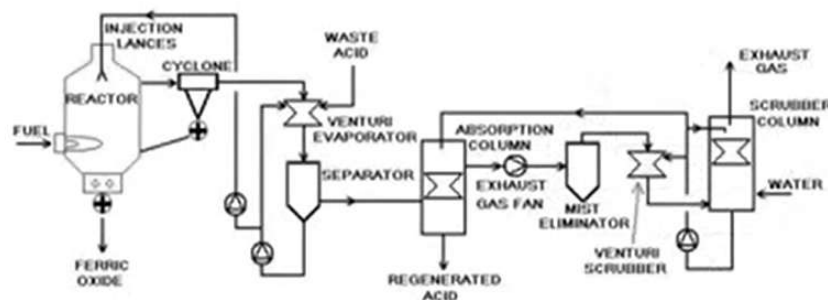
- The course contents are distinctively subdivided into:
- **Control Technologies and PLCs**
 - Role of Control Technologies in Automation
 - Automation through PLCs
 - What are PLCs?
 - Everything related to PLCs





Industrial Automation

- The course contents are distinctively subdivided into:
- **Industrial Communications**
 - Communication Networks
 - Documentation
 - Quality Control





History

- One of the aspect of living in a society is to share knowledge and skill.
- As societies began to develop, need for things ranging from knives to clothes, and from food to weapons, rose quickly.
- The work *manufacturing* was derived from two Latin words, *manus* (hand) and *factus* (make), so that the combination means *made by hand*.



History

- Initially and for a long long time, things were hand made with a little use of hand tools.
- In the 18th century, the Britons changed the way things were manufactured and introduced factories and machines.
- This was the first industrial revolution and it centered on iron, coal and textile products.
- **Major product:** Steam Engine





History

- In late 19th and early 20th century Soviet Union joined the race that forced Britons to expand industries into eastern regions like China and India.
- This was the 2nd Industrial revolution when the industry began to exploit many natural and synthetic resources like lighter metals, alloys, plastics, gas, oil and electricity etc.
- **Major product:** Steel, Chemical synthesis, telegraph and telephone





History

- After mid-20th century the Nuclear energy was exploited, this was the 3rd Industrial Revolution.
- Moreover, advances in electronics, telecommunications and computers totally revamped the industrial setup.
- **Major products:** Programmable controllers, Robots (fully automated machines)



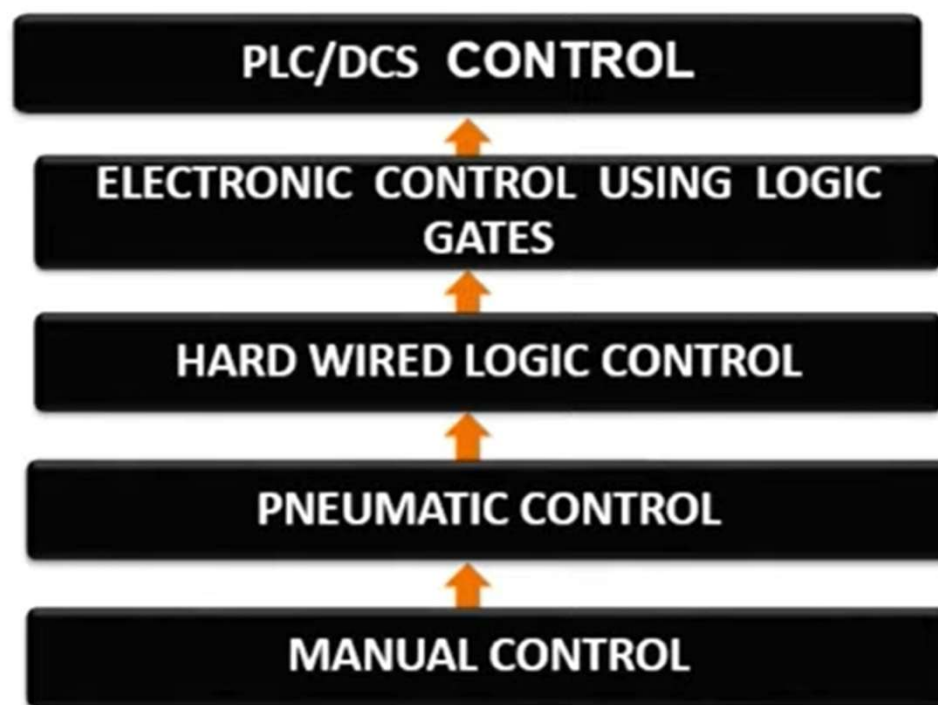


History

- Early 21st century, that is right now we are experiencing 4th Industrial Revolution.
- The 3rd revolution gave us enough fuel to upgrade every aspect of the technology.
- **Major product:** Internet and communication technologies (4G, 5G etc.)



HISTORY OF AUTOMATION



Introduction

Automation is a set of technologies that results in operation of machines and systems without significant human intervention and achieves performance superior to manual operation.

‘Automation’ is derived from Greek words

Auto - self

Matos - moving.

Automation therefore is the mechanism for systems that “move by itself”

Effects of automation

- Increased productivity - Production can run round the clock, except for a few maintenance interval periods
- Improved quality
- Reduced operation time and work handling time
- Reduced direct human labor costs and expenses
- Reduced processing time - larger quantities can be shipped faster
- Automation relieves people of boring, physically heavy or hazardous work

Types of Automation

- 1.Home Automation
- 2.Building Automation
- 3.Industrial Automation
- 4.Process Automation
- 5.Public Transport Automation
- 6.Scientific Automation

Industrial Automation

FIELD INSTRUMENTS

- Level Sensor, temperature Sensor,
- Pressure Sensor, Flow Sensor, load Sensor etc.

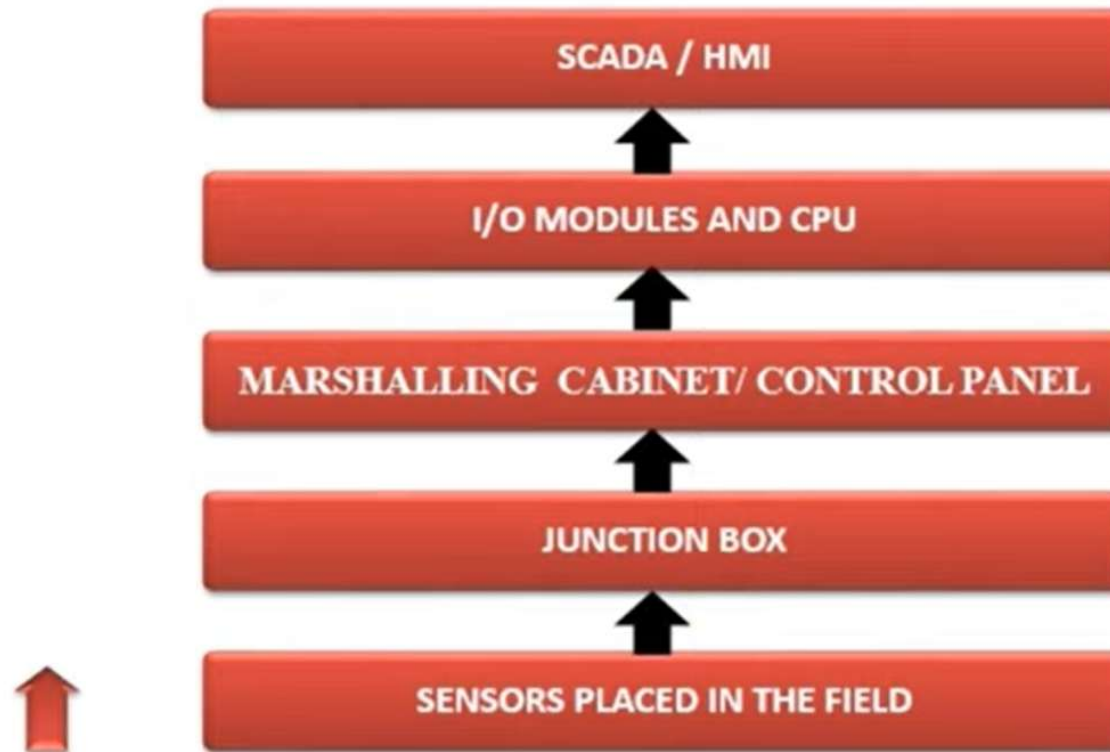
CONTROL HARDWARE-(HW+SW)

- PLC (Programmable Logic Controller)
- DCS (Distributed Control System)
- PID (Proportional Integral Derivative Controller)
- CNC (Computer Numeric Controller)

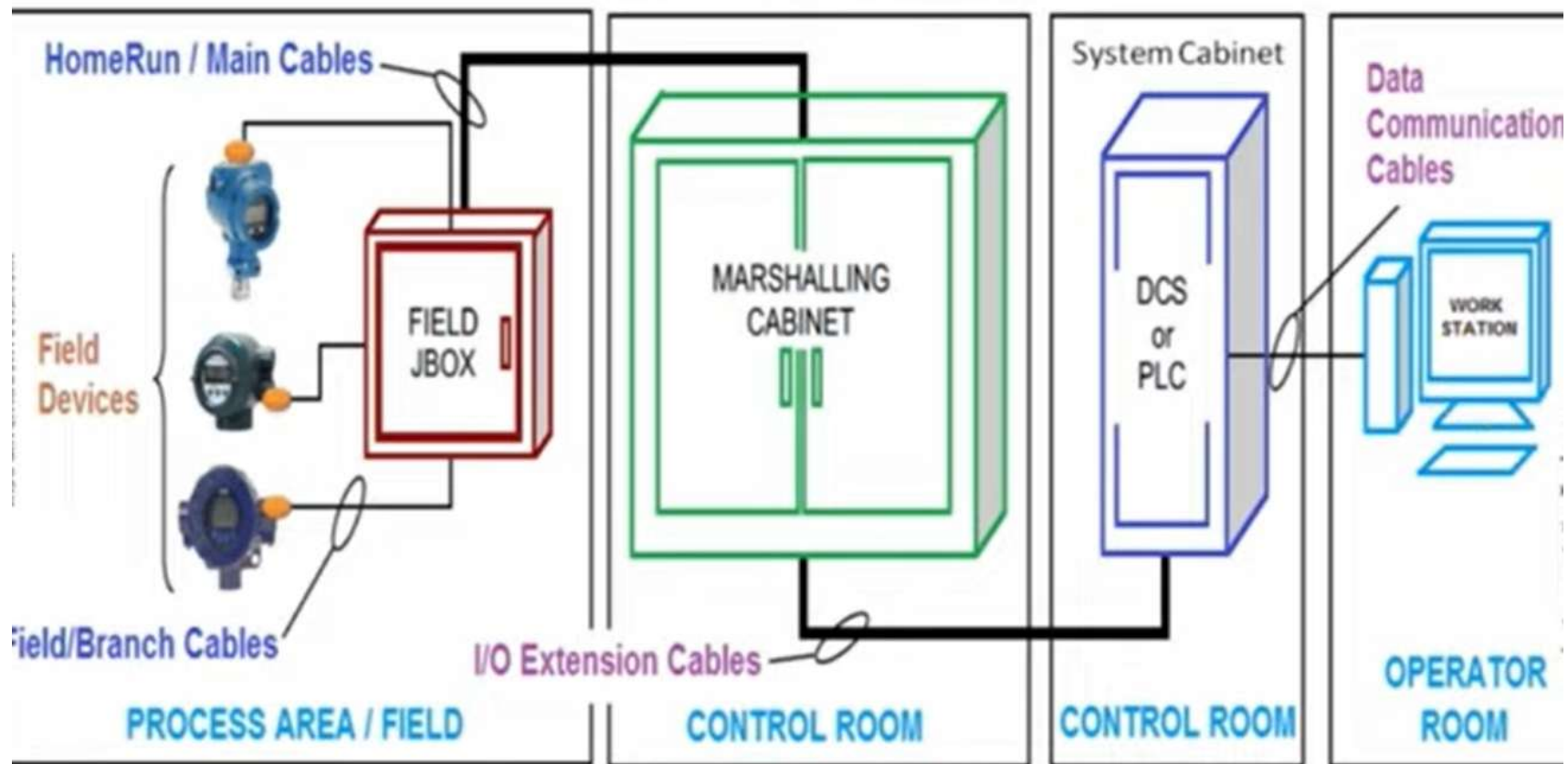
CONTROL SOFTWARE-(SW)

- SCADA (Supervisory Control and Data Acquisition)

AUTOMATION INSTALLATION



Control System Basic Wiring Practice



CYBERSECURITY IN INDUSTRIAL AUTOMATION?

Why should we care about the **security of industrial automation and control systems?**

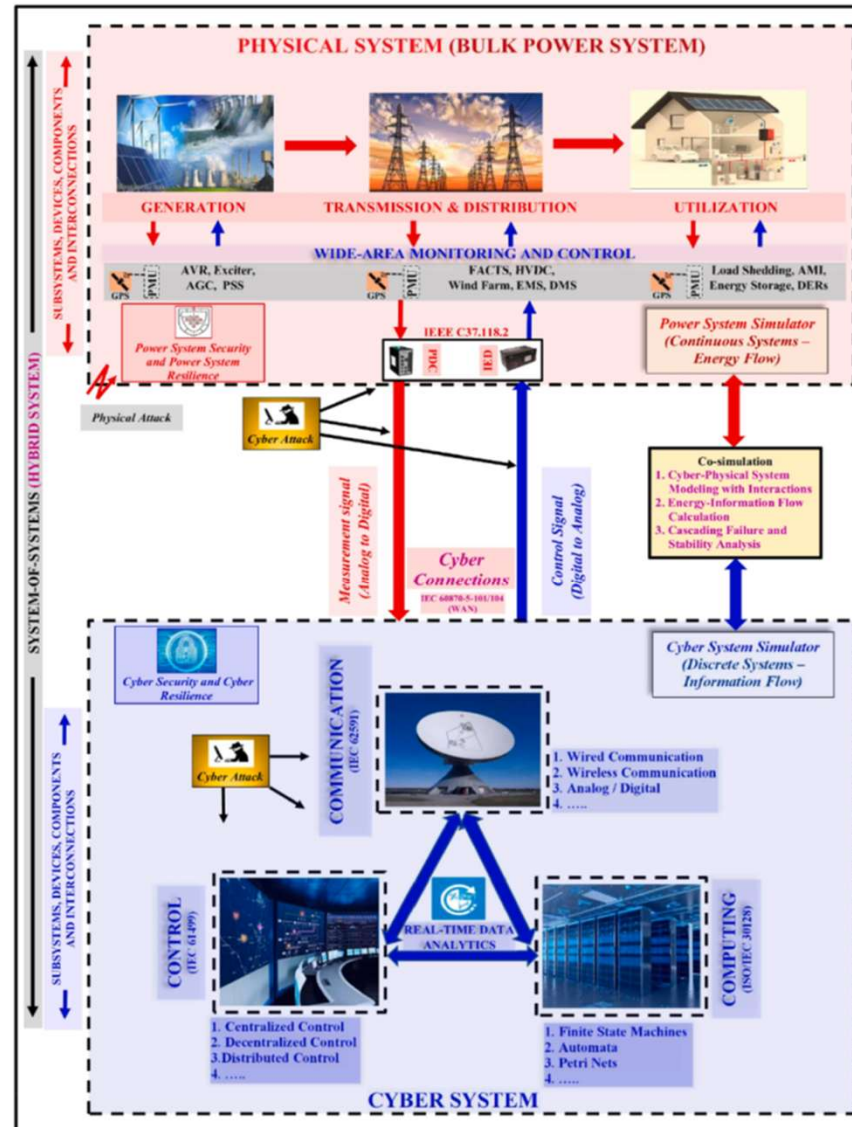
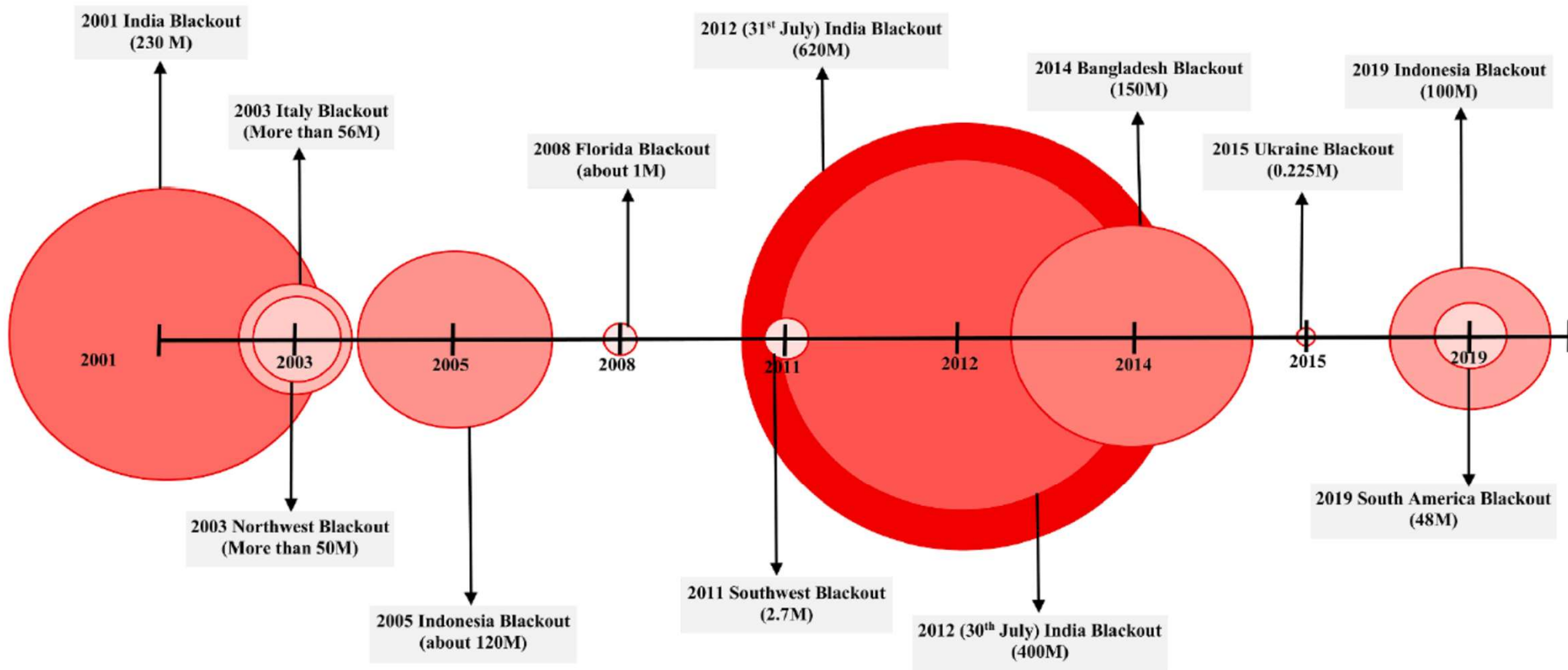


Fig. 1. A Conceptual View of the Cyber-Physical Power System (CPPS).

Major Cyber-Physical Attacks In The Energy Industry Sector.

Year	Location	Attack Objects	Attack Type	Impact
1982	Russia	Gas pipeline control system software	Code Manipulation	3 kilotons TNT equivalent explosion
1999	Bellingham, USA	Slowdown of SCADA system of a gasoline pipeline	Code Manipulation	Huge fireball that killed 3 people and injured many others
2003	Ohio, USA	Slammer worm penetrated the nuclear plant control system	Malware Injection	Parameter display system was off for 5 h
2007	Idaho National Laboratory, USA	Aurora attack manipulated a circuit breaker of a diesel generator	False Data Injection	Exploded generator
2008	Turkey	Attackers manipulated control system parameters of the oil pipeline	False Data Injection	Oil explosion and 30 k barrels are spilled in water
2010	Iran	Stuxnet worm penetrated the nuclear power plant SCADA system	Malware Injection	At least 14 industrial locations in Iran were infected, including a uranium enrichment plant.
2012	Saudi Arabia & Qatar	Malware affected Aramco and RasGas	Malware Injection	Generation and delivery of energy have been affected.
2015	Ukraine	Attack on the breaker's settings in 3 distribution companies	False Data Injection	For a few hours, 225 k customers were without service.
2016	Ukraine	Malware industroyer intrudes into transmission substation control systems	Malware Injection	It amounted to a one-fifth reduction in power use at that time of night.
2017	Saudi Arabia Oil Refinery	Malware Triton affected safety systems of a power station	Malware Injection	It triggers the plant's emergency shutdown system to go into action.
2019	Power Utilities @ California & Wyoming, USA	Communication network bombarded with network traffic	Denial of Service	For a short time, electrical system operations would be disrupted, but this would not result in a blackout.
2019	Kudankulam Nuclear Power Plant, Tamil Nadu, India	Virus having infected the systems at the power plant	Man-In-The Middle attack	The infected systems were segregated from the vital internal network, and the rest of the system's functionality was unaffected.



Examples of extreme cyber-physical attacks on the energy sector, where M signifies the number of customers without electricity in millions.

IACS **Security**: A Component of a Nation's Critical Infrastructure

- As defined by ANSI/ISA-99.00.01, **industrial automation and control systems (IACS) include** (but are not limited to) distributed control systems (**DCSs**), programmable logic controllers (**PLCs**), remote terminal units (**RTUs**), intelligent electronic devices, supervisory control and data acquisition (**SCADA**) systems, networked electronic sensing and control, and monitoring and diagnostic systems.
- There is also an emerging trend in many organizations toward consolidating some **overlapping activities in IACS and corporate IT systems**. This trend is motivated by the cost savings achievable by avoiding the use of **disparate platforms, networks, software, and maintenance** tools and by an increased capability to run the total organization more efficiently and effectively.

- An **important issue associated with the merging of these two systems** is that, in many cases, both IACS and corporate IT environments use the **same security model**.
- This overlap introduces the possibility of the corporate Internet connection exposing critical operations to **additional threats** and compromising the real-time, deterministic requirements of plant control systems.

ANSI/ISA-99.00.01 states that an **industrial automation and control system** is:

A **collection of personnel, hardware, and software** that can affect or influence the **safe, secure, and reliable operation of an industrial process**. These systems include, but are not limited to:

- a. industrial control systems, including distributed control systems (DCSs), programmable logic controllers (PLCs), remote terminal units (RTUs), intelligent electronic devices, supervisory control and data acquisition (SCADA), networked electronic sensing and control, and monitoring and diagnostic systems. (In this context, process control systems include basic process control system and safety instrumented system [SIS] functions, whether they are physically separate or integrated.)

b. associated information systems, such as advanced or multivariable control, online optimizers, dedicated equipment monitors, graphical interfaces, process historians, manufacturing execution systems, and plant information management systems.

c. associated internal, human, network, or machine interfaces used to provide control, safety, and manufacturing operations functionality to continuous, batch, discrete, and other processes.

SCADA Systems

ANSI/ISA-99.00.01 defines a SCADA system as “a type of loosely coupled distributed monitoring and control system commonly associated with electric power transmission and distribution systems, oil and gas pipelines, and water and sewage systems.”

A SCADA system comprises both hardware and software, and the classical SCADA system model includes the following components:

- Human-machine interface (HMI)
- Master terminal unit (MTU)
- Remote terminal unit (RTU)
- Programmable logic controller (PLC)
- Intelligent electronic device (IED)

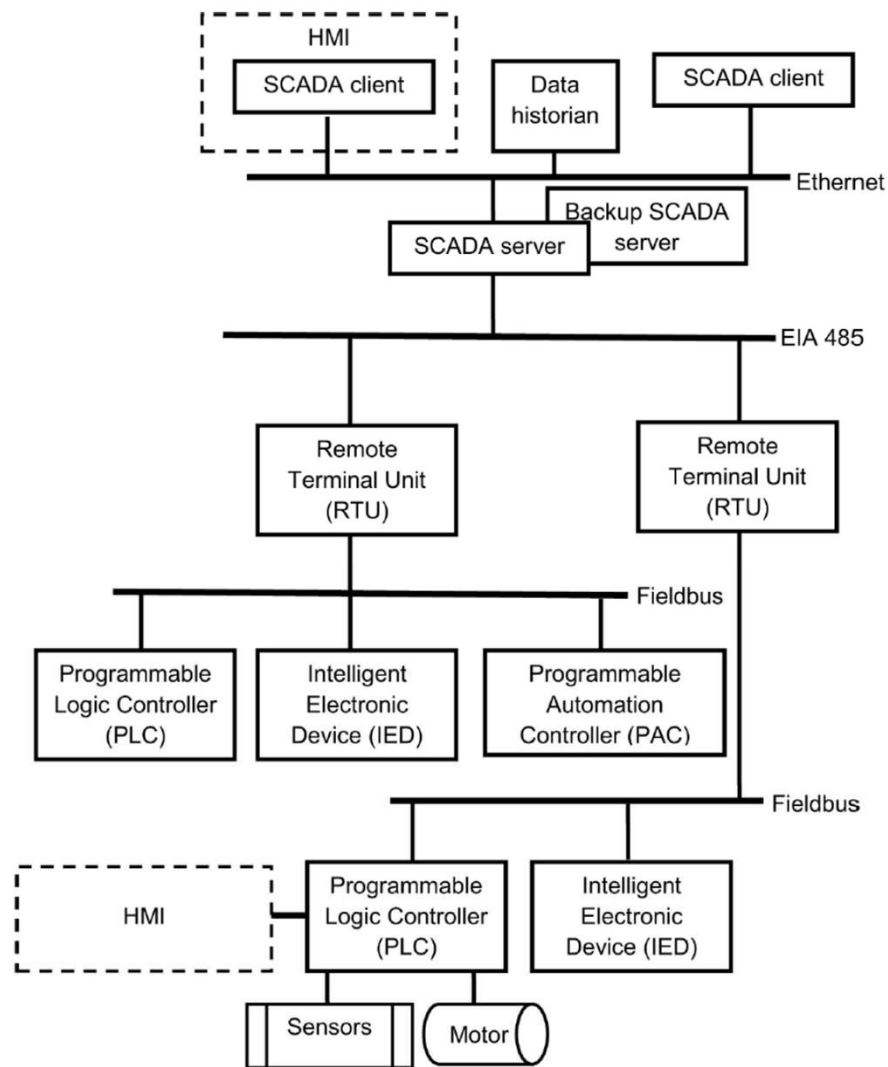


Figure 1-1. A client-server SCADA architecture example

Distributed Control Systems

- A DCS includes a **supervisory controller** running on a control server and a number of **distributed controllers**.
- The **supervisory controller** transmits data set points to the remote controllers and acquires data from them.
- The **distributed controllers** control process elements by communicating with them over a field-bus-type network based on the information received from the supervisor.

ANSI/ISA-99.00.01-2007⁵ classifies a distributed control system as:

A type of control system in which the system elements are dispersed but operated in a coupled manner.

NIST 800-82⁶ defines a distributed control system as “control achieved by intelligence that is distributed about the process to be controlled, rather than by a centrally located single unit.”

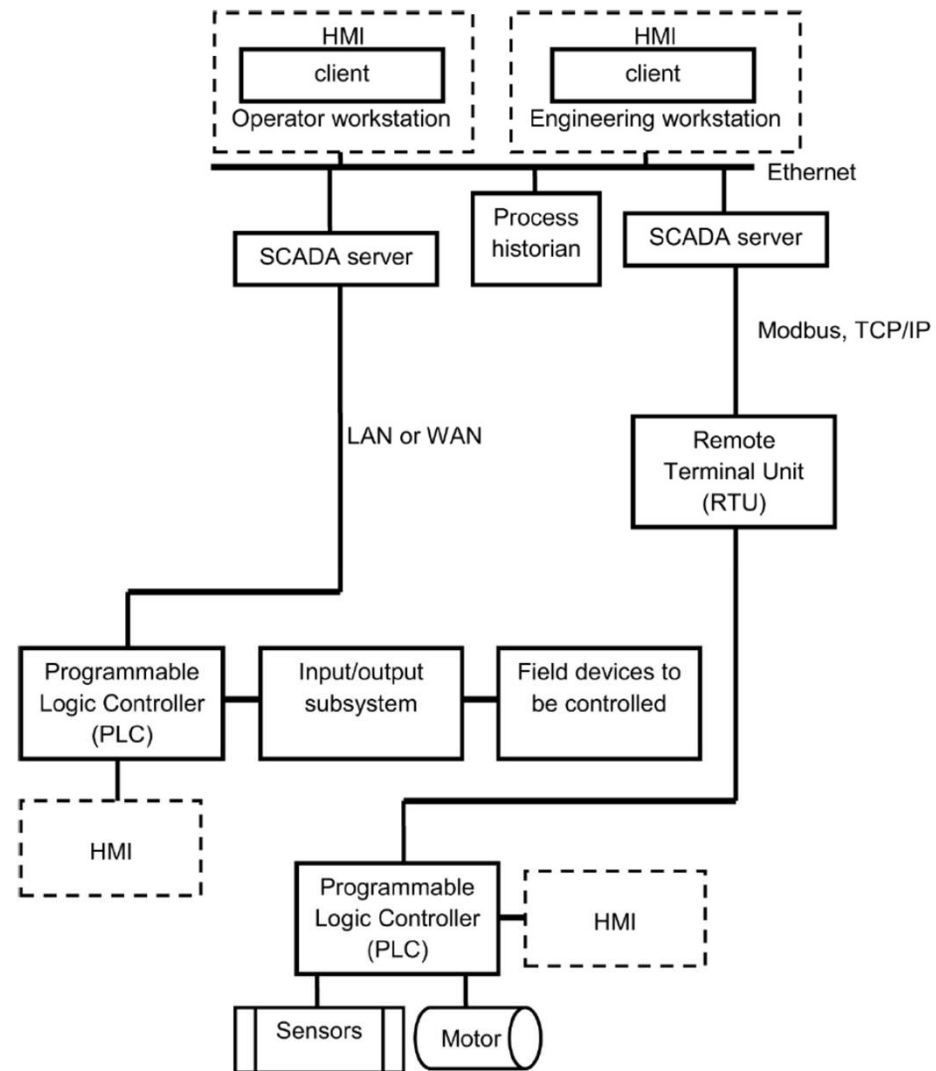
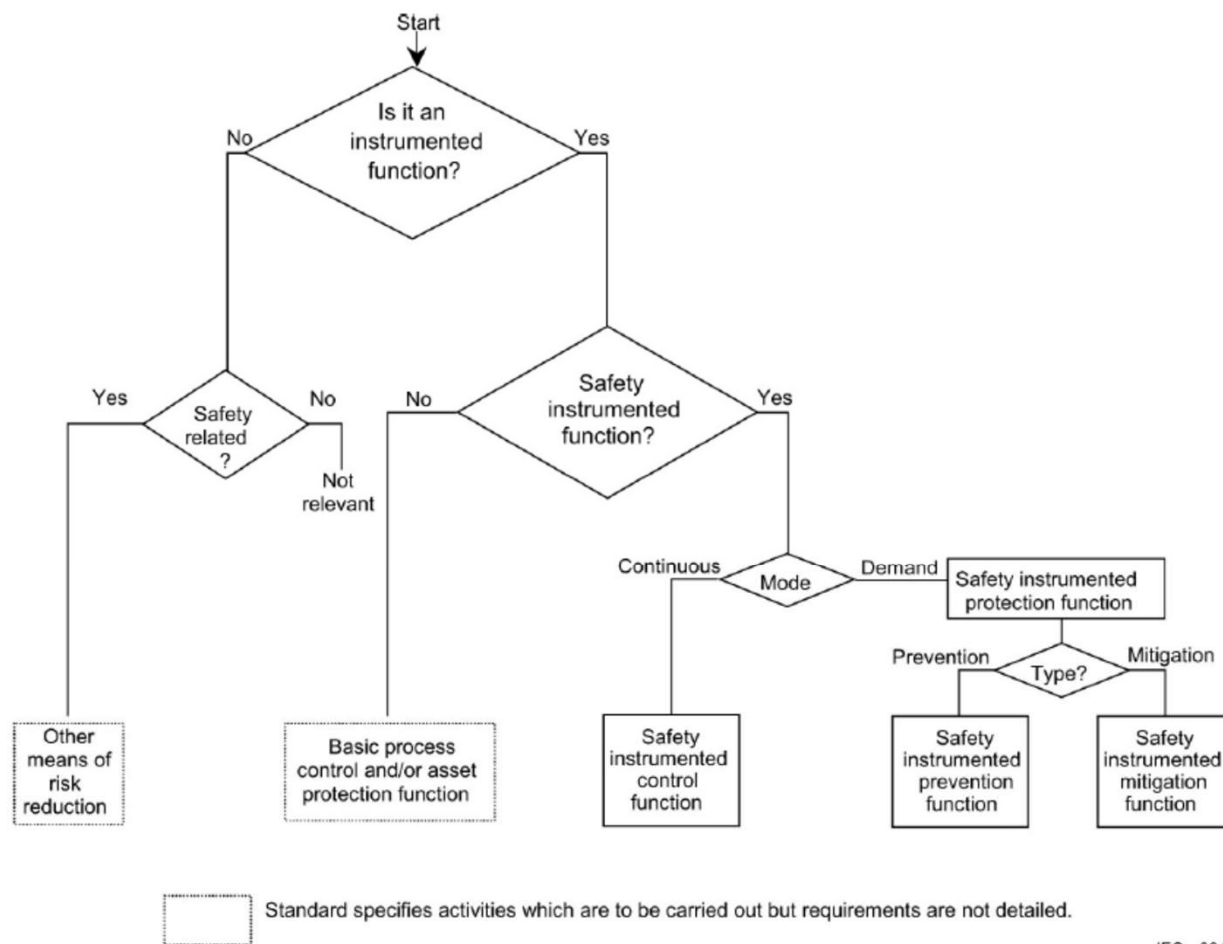


Figure 1-2. Example of a distributed control system

Safety Instrumented Systems

- Safety instrumented systems (SISs) have been widely used in the process industry to maintain a process in a safe condition during a hazardous situation, for example, if critical set points are exceeded or safe operating conditions are breached. SISs are sometimes referred to as safety shutdown (SSD) systems or emergency shutdown (ESD) systems.
- ANSI/ISA-84.00.01-2004, Part 1(IEC 61511-1 Mod)7 defines an SIS as an “instrumented system used to implement one or more safety instrumented functions (SIFs). An SIS is composed of any combination of sensor(s), logic solver(s), and final element(s).”

- The standard also defines an SIF as a “safety function with a specified safety integrity level which is necessary to achieve functional safety and which can be either a safety instrumented protection function or a safety instrumented control function.”
- An SIS is usually implemented in parallel with conventional control systems to reduce the risks associated with explosive or other dangerous environments.



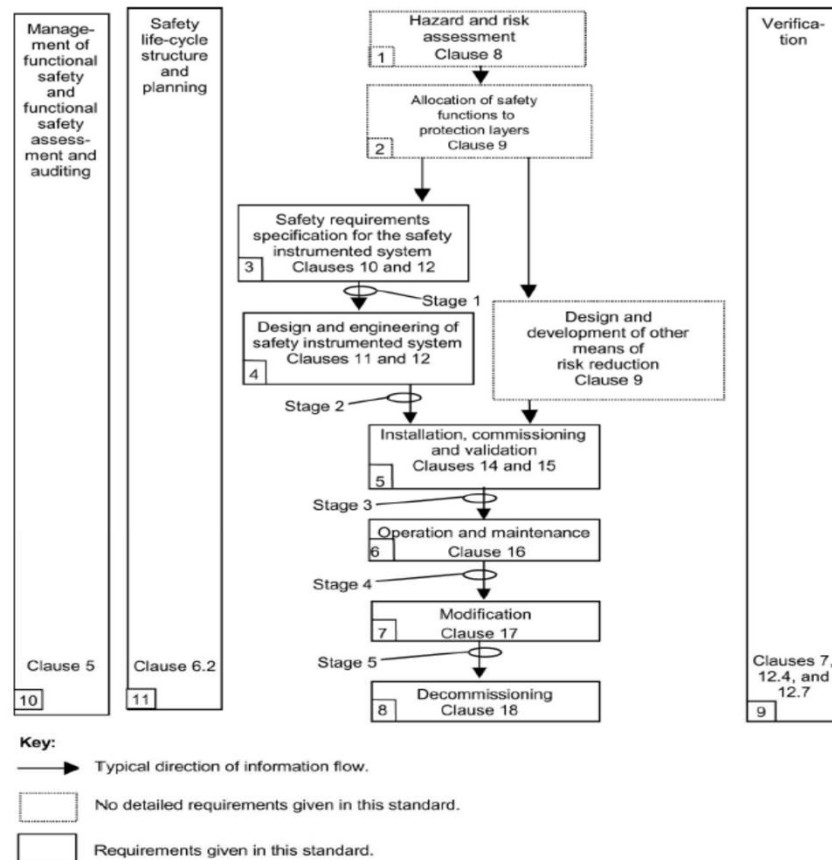
IEC 3243/02

Figure 1-3. Safety instrumented functions vs. other process functions

[Source: ANSI/ISA-84.00.01-2004 Part 1 (IEC 61511-1 Mod)]

Three tiers of an organization-wide risk management paradigm:

1. Tier 1 – Organization: Establishes and implements governance structures consistent with the organizational mission and goals.
2. Tier 2 – Mission/Business Processes: Designs, develops, and implements mission/business processes to support the mission defined in Tier 1.
3. Tier 3 – Information Systems: Integrates risk management activities into the system development life cycle (SDLC) of organizational information systems and addresses the resilience of organizational information systems.



NOTE 1 Stages 1 through 5 inclusive are defined in 5.2.6.1.3.
 NOTE 2 All references are to Part 1 unless otherwise noted.

Figure 1-4. SIS life cycle and functional safety elements
 [Source: ANSI/ISA-84.00.01-2004 Part 1 (IEC 61511-1 Mod)]

Industrial Automation and Control System Protocol Summary

- A protocol defines the rules for entities to use in communicating with each other.
- In order to better manage communications and compartmentalize the activities required to establish, maintain, use, and close communication links, high-level models are used.
- In particular, layered architectures are useful in defining the various communication functions in a hierarchical manner.
- Two widely popular implementations of the layered Architecture paradigm are the Open Systems Interconnect (OSI) model and the Transmission Control Protocol/Internet Protocol (TCP/IP) model.

OSI Model

- In the early 1980s, the [International Organization for Standardization \(ISO\)](#) Developed the Open Systems Interconnection (OSI) reference model.
- The model functions [by encapsulating data and transferring it to an adjacent lower level](#), where the data is again encapsulated by the lower level, for example by adding a header.
- [Table 1-1 summarizes the OSI model layers and the protocols associated with each layer.](#)

Table 1-1. OSI Model Layers

Layer	Role	Protocols
7-Application	Provides for application support and verifies receiver availability.	SMTP (Simple Mail Transport Protocol), FTP (File Transfer Protocol), SNMP (Simple Network Management Protocol)
6-Presentation	Conducts translations and conversions for the application layer, including compression, decryption, and encryption.	MPEG (Motion Pictures Experts Group), HTTP (Hypertext Transfer Protocol), JPEG (Joint Photographic Experts Group)
5-Session	Sets up communication connections, controls the data transmission during a session, and terminates the connection.	NFS (Network File System) - Supports file sharing, RPC (Remote Procedure Call)
4-Transport	Sets up a logical connection between the transmitter and receiver. The transport layer is connection-oriented and ensures data delivery to the receiver.	TCP (Transmission Control Protocol), UDP (User Datagram Protocol)

3-Network	Performs error checking, addressing, routing, and node traffic control.	IP (Internet Protocol), ICMP (Internet Control Message Protocol)
2-Data Link	Includes the media access control (MAC) sublayer and logical link control sublayer. Performs error-free packet transmission.	PPP (Point-to-Point Protocol), ARP (Address Resolution Protocol)
1-Physical	Converts packets into electrical or optical signals for sending on the transmission media, such as wiring or cables. Defines network electrical and mechanical interfaces to the network transmission media.	EIA-232C (RS-232C), EIA-422-B (RS-422)

TCP/IP Model

- In the 1970s, the U.S. Department of Defense developed the Transmission Control/Internet Protocol (TCP/IP), which is the basic protocol of the Internet.
- It was developed prior to the OSI model and is named from two of its most important protocols: the Transmission Control Protocol and the Internet Protocol.
- The OSI model is a more formalized layered suite of protocols, but inherited some its architecture from TCP/IP.
- Table 1-2 summarizes the four TCP/IP layers and the protocols associated with each layer.

Table 1-2. TCP/IP Model Layers

Layer	Role	Protocols
4-Application	Provides the interface to the user.	Telnet, FTP, SNMP, SMTP
3-Host-to-Host or Transport	Provides connection-oriented end-to-end communications, supports error-free packet delivery, and organizes data in to packets.	TCP, UDP
2-Internet	Performs addressing and routing functions.	IP, ARP, ICMP PPP
1-Network Access	Combines the operations of the OSI physical and data link layers	EIA-422-B (RS-422), EIA-232C (RS-232C), IEEE 802.2 Logical Link Control

OPC (Object Linking and Embedding [OLE] for Process Control)

Open Platform Communications (OPC) is an interoperability standard for the secure exchange of industrial automation data. It is designed to be platform-dependent so devices from different vendors can exchange information.

NIST 800-82⁹ defines OPC as “a set of open standards developed to promote interoperability between disparate field devices, automation/control, and business systems.” The OPC standards specifications are now known as the Data Access Specification. OPC was built on the OLE COM (Component Object Model) and DCOM (Distributed Component Object Model) technologies of Microsoft, which specified a standard set of objects, interfaces, and methods for use in process control and manufacturing automation applications. Because of some vulnerabilities in Microsoft’s software, OPC should only be applied between the demilitarized zone (DMZ) and control networks and not used between the corporate network and the DMZ.

The following list summarizes some of the popular OPC specifications:

- OPC Alarms & Events – Supports process alarms, messaging, and event notifications on demand
- OPC Batch – Provides OPC functionality to batch processes
- OPC Command – Provides interfaces that support OPC clients and servers in managing commands sent to devices
- OPC Data Access – Provides for the transmission of data from PLCs, IEDs, and DCSs to HMI clients
- OPC Data eXchange – Supports interoperability among a variety of vendors and communication and management services among Ethernet fieldbus connections

- OPC Historical Data Access – Provides archiving of historical data
- OPC Security – Safeguards critical data through access control and integrity protection mechanisms
- OPC Unified Architecture – Defines specifications that do not use Microsoft COM in order to enhance cross-platform usage

OPC Unified Architecture

- OPC Unified Architecture (OPC UA) is an OPC Foundation Web services technology designed to provide the functions of OPC, but to support interoperability among different platforms and achieve independence from Microsoft COM/DCOM.
- It can also interface easily with corporate enterprise systems and manufacturing management systems. Some of the features of OPC UA are:
 - Integrated security
 - Multiplatform compatibility
 - Scalable
 - Object-oriented
 - Ability to identify specific components
 - Information modeling, including how data is communicated

Modbus/TCP Model

- Modbus is an OSI Layer 7 serial communication protocol that was developed by Modicon in 1979 for use with data acquisition and control systems.
- Modbus enables reliable communication among various control devices, including PLCs.

DNP3 Protocol

- The **Distributed Network Protocol (DNP)3** is an open SCADA protocol that is used for serial or TCP/IP communications between control devices.
- It is widely used by utilities, such as **water companies and electricity suppliers**, for the exchange of data and control instructions between master control stations and remote computers or controllers called outstations.
- **Figure 1-5** is an example of a **SCADA application, using DNP3** with a data concentrator in a hierarchical mode.

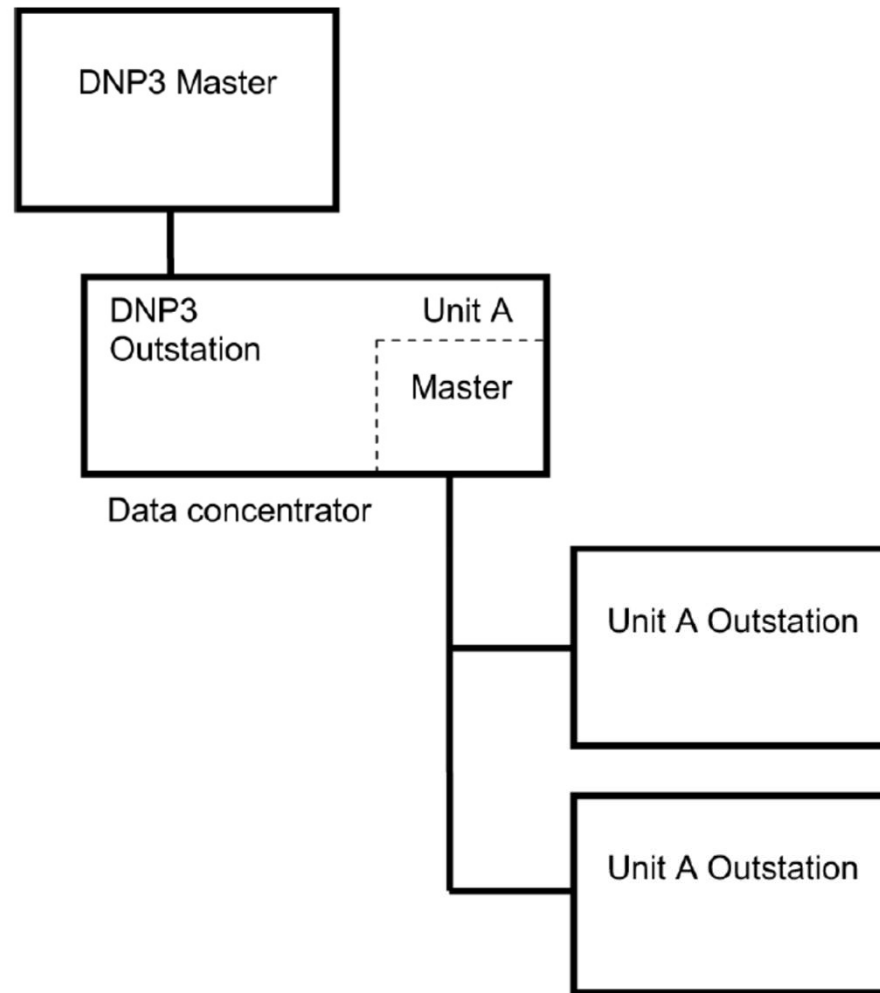


Figure 1-5. DNP3 example usage

Utility Communications Architecture 2.0/IEC61850

- The [Utility Communications Architecture](#) (UCA), version 2.0, comprises a set of communications protocols [for use by electric utilities](#).
- In 1999, UCA 2.0 evolved to the layered architecture [standard IEC 61850 for substation automation](#).
- [IEC 61850-enabled IEDs to obtain power grid condition data](#) via an Ethernet process bus and from merge units, which provide interfaces to field devices.
- Legacy devices utilize an IEC 61850 wrapper around legacy protocols for compatibility. [IEC 61850 includes the following features:](#)

- Abstract definitions of services and data
- Definitions for IED communications
- IED communications through substation buses
- Mapping of services to protocols
- Object-oriented approach
- Standardized configuration language (SCL)

Figure 1-6 provides an overview of an IEC 61850-based substation.

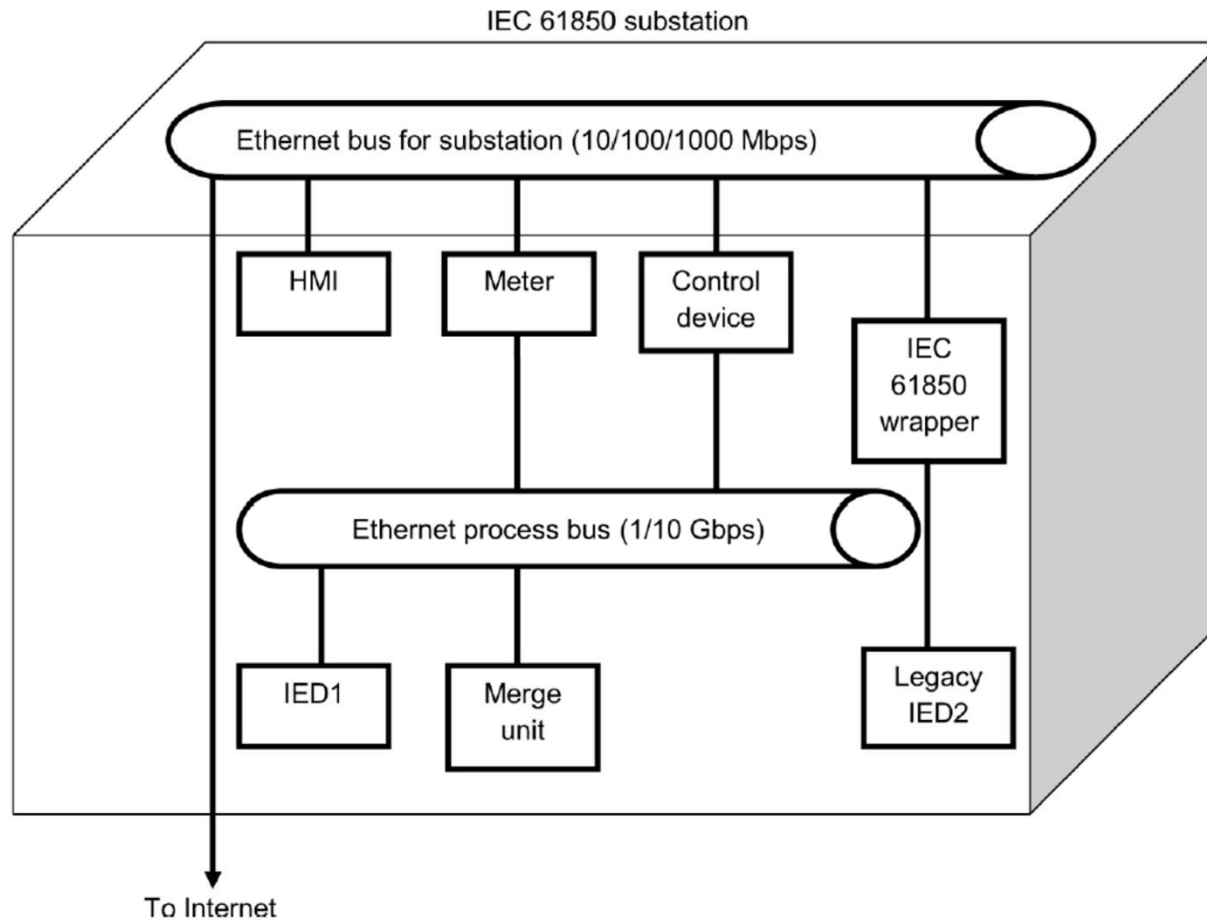


Figure 1-6. An IEC 61850 substation

Profibus

- Process fieldbus or Profibus is an open standard that defines the electrical, mechanical, and functional aspects of a fieldbus that addresses deterministic data acquisition implementations.
- It is compatible with European international fieldbus standard EN 50 170.
- There are **three versions of Profibus**: Profibus Process Automation (PA), Profibus Decentralized Peripherals (DP), and Profibus Fieldbus Message Specification (FMS).

- Profibus PA communicates with control devices and data acquisition components through a common serial bus and is specifically designed for use in hazardous and explosive environments through current limiting elements.
- Profibus DP communicates over a high-speed network with sensors and actuators through a central controller and is suited for Factory automation requirements.

Controller Area Network

- Controller Area Network (CAN) protocols, also known as ISO Standard 11898-1, were developed in 1986 by Robert Bosch GMBH in Germany for control applications in automobile production lines.
- The protocol uses an altered form of Ethernet Carrier Sense Multiple Access with Collision Detection (CSMA/CD) to implement a deterministic protocol for SCADA systems.

EtherNet/IP

- EtherNet/IP is an application layer protocol designed for industrial automation and control systems applications and is a member of a family of networks that implement the Common Industrial Protocol (CIP™) at its upper layers.
- EtherNet/IP provides control, configuration, and data exchange capabilities on an Ethernet network. A typical EtherNet/IP architecture is shown in Figure 1-7.

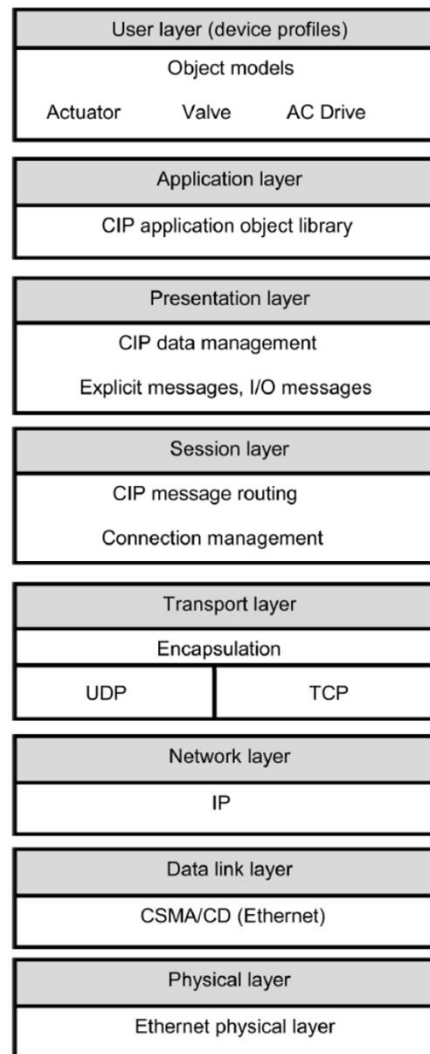


Figure 1-7. A typical EtherNet/IP architecture

openSAFETY Protocol

- OpenSAFETY¹¹ is an application layer protocol sponsored by the Ethernet Powerlink Standardization Group¹² (EPSG), which supports the communication of safety-related data over a variety of industrial networks and ensures data transfer integrity.
- An important characteristic of openSAFETY is that it employs a variable-length frame with a uniform format that supports safe distribution of values.
- To meet these requirements, openSAFETY verifies parameters using cyclic redundancy check (CRC) and encapsulates its messages, which independently tunnel through the extant protocols being used in the data acquisition and control network.
- Figure 1-8 illustrates the relationship of openSAFETY to the other protocols in the OSI layered architecture.

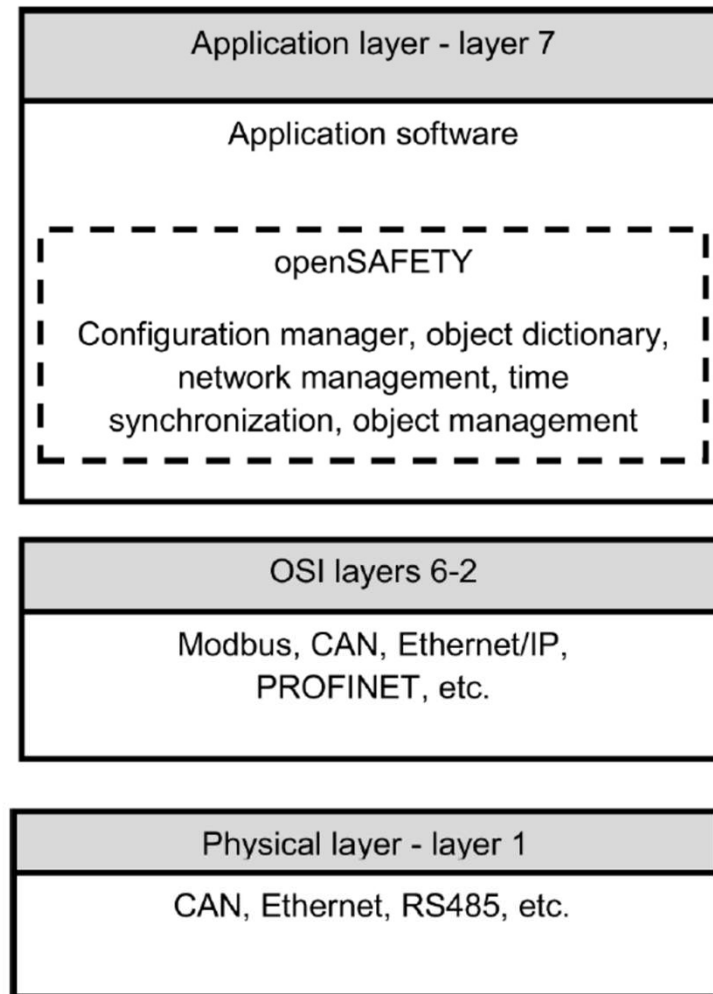


Figure 1-8. A typical openSAFETY architecture

Issues in Industrial Automation and Control Systems Security

No one questions the need to protect a nation's critical infrastructure and essential production processes. The available means and methods required to accomplish this defense are varied and have multiple characteristics associated with them. These characteristics include:

- Hardware cost
- Software cost
- Reliability
- Level of assurance
- The effect of security controls on performance
- Maintenance issues
- Safety issues
- Effect on product quality
- Response to emergency situations
- Interfaces to management systems
- Compatibility with 24/7 operations

These issues and others will be explored in this subject to provide a foundation for securing industrial automation and control systems.

Summary

- Over the past two decades, methodologies and [best practices in the field of Information system security](#) have been developed, refined, and codified.
- They comprise a rich compendium of proven and effective measures to [secure computer systems and networks](#).
- The fundamentals of these developments will be presented in [next chapter as a basis for the evolution of security technologies applicable to industrial automation and control systems](#).
- A number of these paradigms are [not directly applicable](#) to [industrial automation and control systems](#), but many are, and many can be employed with modifications tailored to the unique characteristics of these systems.

- These adapted methods can be effective in securing industrial automation and control systems against threats, such as **viruses, Trojan horses, worms, and attacks** that result in the unauthorized modification and manipulation of critical data, denial of service and availability, unauthorized modification of audit logs, and general disruption of critical operations.
- The information system security approaches embrace a wide variety of disciplines and include:

- Policies, procedures, and standards
- Access control mechanisms
- Physical security
- Identification and authentication technologies, such as biometrics and security tokens
- Audit trails
- Real-time monitoring of system and network information
- Risk management
- Business continuity and disaster recovery planning
- Firewalls to filter message traffic
- Intrusion detection and protection systems
- Public key and secret key cryptography
- Malware detection, isolation, and elimination